# ABB ENGINEEREDX 2.0

PROBLEM STATEMENT-1 *Digital Twin of Transformer and Electrical Equipment*

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|---|---|
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| BATCH | **B.Tech CSE · 2027 batch** |
| PROBLEM STATEMENT | **Digital twin of transformer or any other electrical equipment** |

## Inputs Considered :-

Digital Twin relies on four main type of input data to accurately understand, predict, and simulate how the equipment behaves in real-world conditions.

- Real-time sensor streams
- Static asset data
- Historical operational data
- Domain priors & standards

### 01 Real-time sensor streams

Oil temperature (top& bottom), winding hotspot temperature, primary/secondary voltage and current, Dissolved Gas Analysis ($H_2$, $CH_4$, $C_2H_2$, $C_2H_4$, $C_2H_6$, CO, $CO_2$), vibration spectra, partial discharge counts, OLTC tap position, ambient temperature and humidity.

**Why →** *Live vital signs that drive every prediction the twin makes about the asset's present state.*

### 02 Static asset data

Nameplate ratings (kVA, voltage class, frequency), per-unit impedance, cooling class (ONAN / ONAF / OFAF), insulation class, winding configuration, OEM design specifications, geometric and material parameters from the transformer datasheet.

**Why →** *Calibrates the physics model to this specific asset rather than a generic transformer.*

### 03 Historical operational data

Load profiles across seasons,switching events, overload incidents, past maintenance records, fault history, repair logs, and any prior alarm events. Time-aligned with weather and grid-loading data where available.

**Why →** *Training signal for ML models and baseline distribution for anomaly detection.*

### 04 Domain priors & standards

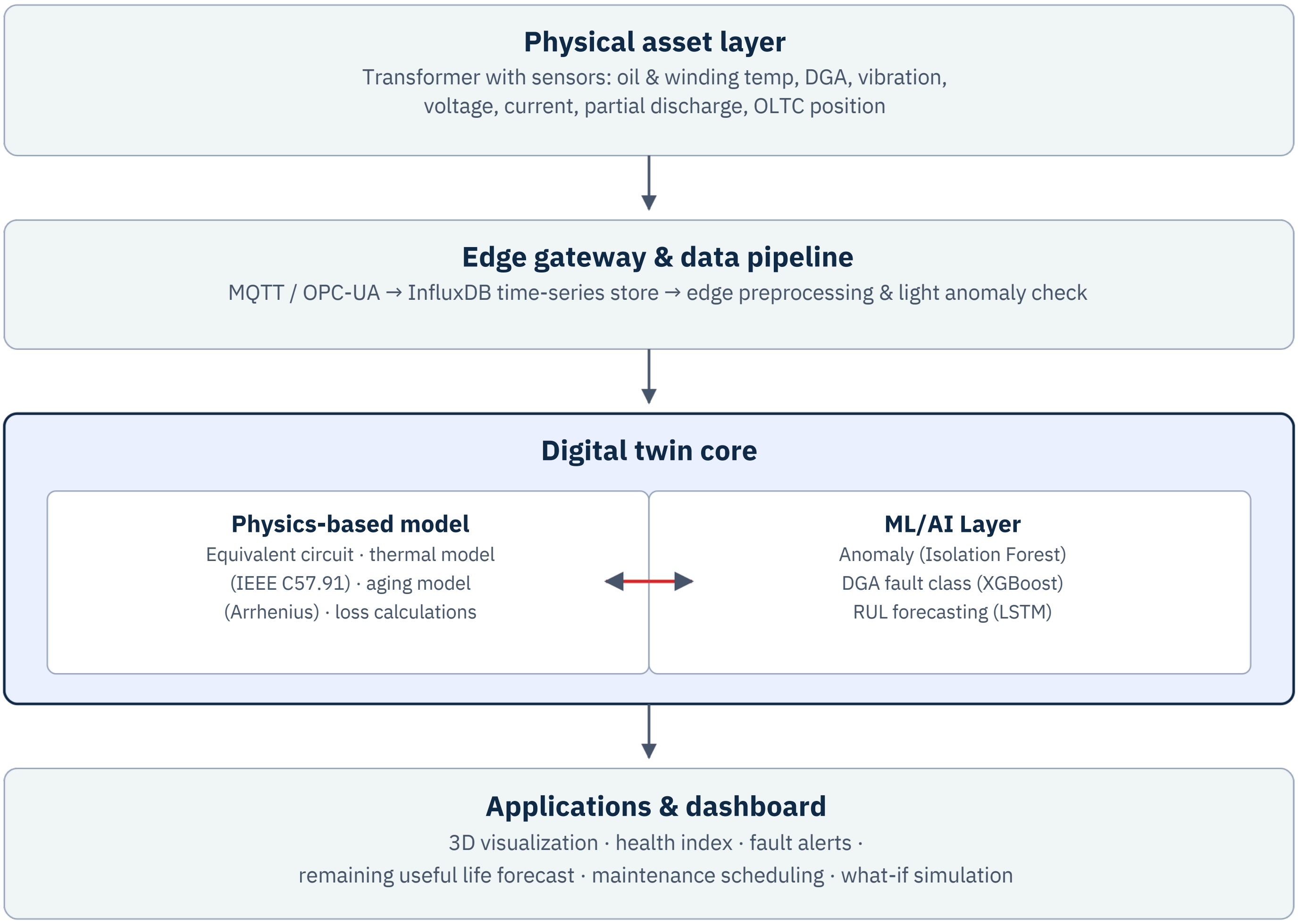
IEEE C57.91 loading guide for hotspot & aging , IEC 60599 for DGA interpretation, the Duval Triangle and Rogers Ratio methods for fault typing, and the Arrhenius equation for paper-insulation thermal aging.

**Why →** *Encodes 70+ years of transformer engineering. Beats purely data-driven models when training data is limited.*

# Process — Architecture

### A four-tier system : -

- Physical asset
- Edge ingestion
- Digital twin core
- Operator-facing applications.




## Data flow

**Upward:** IoT sensors stream measurements through an edge gateway using industrial protocols (MQTT, OPC-UA). Edge nodes do lightweight preprocessing and a first-pass anomaly check before forwarding to the cloud time-series store.

**Twin reasoning:** The digital twin core fuses two complementary engines. The physics model predicts *expected* behavior under current loading and ambient conditions; the ML layer interprets *deviations* from expectation and forecasts future degradation. Neither approach alone is sufficient — together they outperform either.

**Downward:** Insights flow back through dashboards and APIs into operator workflows: real-time alerts, condition-based maintenance schedules, and "what-if" load simulations. The architecture integrates natively with **ABB Ability** on Microsoft Azure Digital Twins.

**TECH STACK-** *MQTT, Azure Digital Twins, OPC-UA, InfluxDB, Python, FastAPI, PyTorch, XGBoost, scikit-learn, Three.js, Plotly, Docker, Kubernetes.*

# Process — Physics-Based Modeling

The Digital Twin uses physics-based modeling to simulate how the transformer is expected to behave under different operating conditions.

Three physics-grounded sub-models compute what the transformer should be doing at every instant. ML detects when reality deviates.

### 1. Thermal model

IEEE C57.91

Predicts the winding hotspot temperature from load, ambient temperature, and cooling state using the IEEE C57.91 loading guide. Output is compared to the measured hotspot reading: persistent deviation indicates cooling-system degradation, blocked oil flow, or incipient insulation failure long before hard thresholds are breached.

$$\theta_h = \theta_a + \Delta\theta_{tor} + \Delta\theta_{hs}$$

**Inputs:** load current, ambient temp, cooling class → **Output:** predicted hotspot $\theta_h$ (°C)

### 2. Aging & remaining-life model

ARRHENIUS

Insulation paper degrades exponentially with temperature. The Arrhenius equation translates the predicted hotspot trajectory into a "loss-of-life per hour" curve. Integrated over operating history, this gives total insulation life consumed and a physics-grounded prior for the ML-based RUL forecast on the next page.

$$Rate = A \cdot \exp( -E_a / R \cdot T )$$

**Inputs:** hotspot history, activation energy → **Output:** life consumed (%), aging acceleration factor

### 3. Electrical equivalent circuit

EQUIVALENT CKT

Astandard T-model with primary/secondary winding resistance, leakage reactance, and magnetizing branch. Used for two purposes: computing real-time losses (copper + iron) and simulating "what-if" fault scenarios — inter-turn short, ground fault, magnetizing in rush — without touching the physical asset. Modeled in Python with SciPy ODE solvers; can be ported to MATLAB/Simulink or Modelica.

**Inputs:** primary V/I, secondary V/I, impedance parameters → **Output:** losses, fault simulation traces, efficiency map

### Validation strategy

Models are calibrated against the OEM datasheet and refined using historical sensor data via least-squares fitting. Where the physics model and measurements disagree consistently, the parameters are re-estimated. This keeps the "expected behavior" baseline accurate as the asset ages and operating regimes shift over time.

# Process — ML/AI Layer

*Three complementary ML models interpret what the physics model can't.*

- *Pattern-rich anomalies*
- *Fault categories*
- *Time-to-maintenance.*

### Anomaly detection

| | |
|---|---|
| **INPUT** | Multivariate sensor vector + physics residuals (24-dim) |
| **MODEL** | Isolation Forest + LSTM autoencoder (ensemble) |
| **OUTPUT** | Anomaly score ∈ [0,1] every 60s |
| **USE** | First-line alert — "this state is unusual given current loading" |

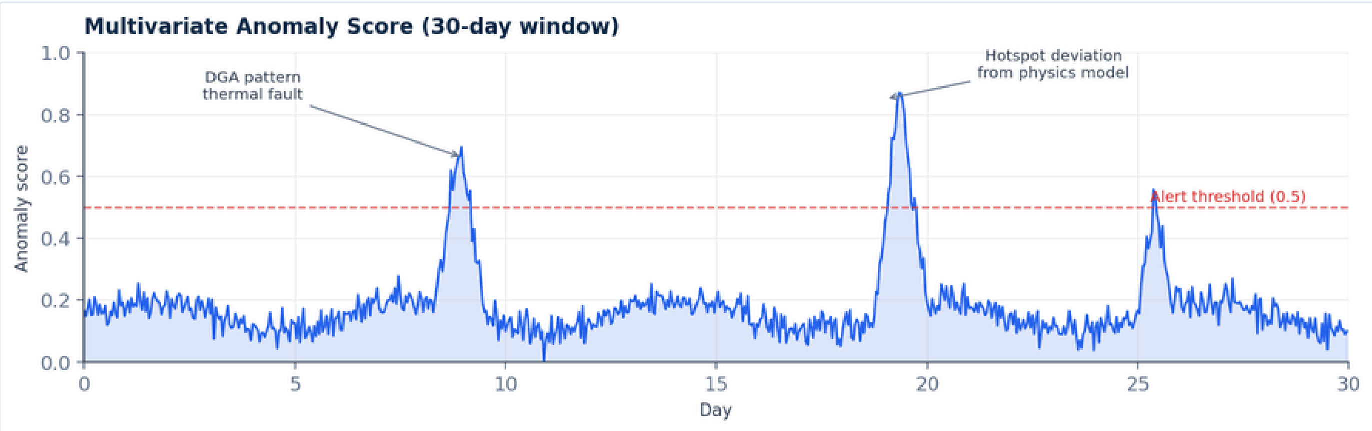
### Fault classification

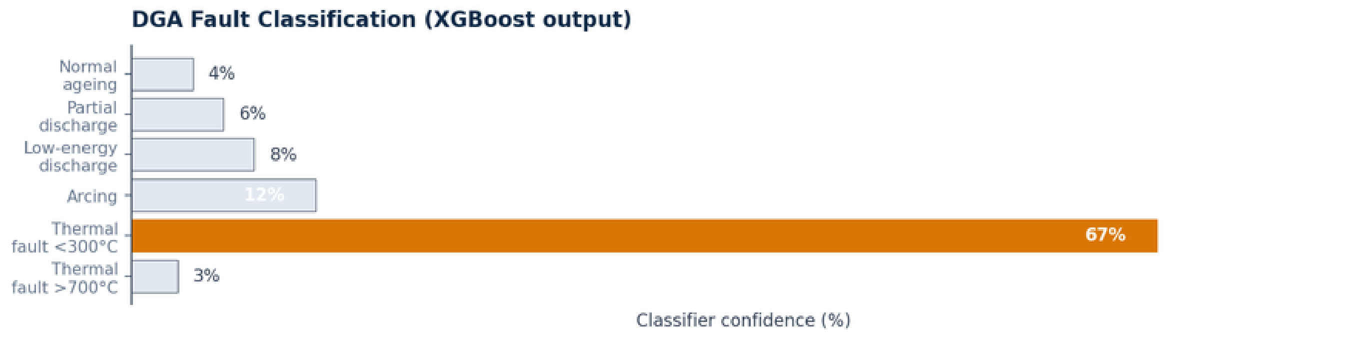
| | |
|---|---|
| **INPUT** | DGA gas ratios + Duval Triangle coordinates |
| **MODEL** | XGBoost classifier (6-class) |
| **OUTPUT** | Fault type + confidence + SHAP explanations |
| **USE** | Diagnose root cause: thermal, PD, arcing, or normal aging |

### RUL forecasting

| | |
|---|---|
| **INPUT** | Time-series of degradation indicators (90-day window) |
| **MODEL** | LSTM with attention / Temporal Fusion Transformer |
| **OUTPUT** | Remaining useful life in days + prediction interval |
| **USE** | Drives condition-based maintenance scheduling |

**THE HYBRID LOOP** The physics model predicts expected behavior under current loading. The ML layer interprets the *residual* — the gap between expected and observed, to flag faults and forecast life. Neither approach alone matches this fidelity: pure physics misses unmodeled degradation modes, while pure ML demands prohibitive training data. The combination is how industrial-grade digital twins actually work.



# Expected Output

*A live operator dashboard plus quantified business outcomes. Architecture extends beyond transformers.*

**Transformer Health Dashboard — Asset TR-04 / 132 kV**

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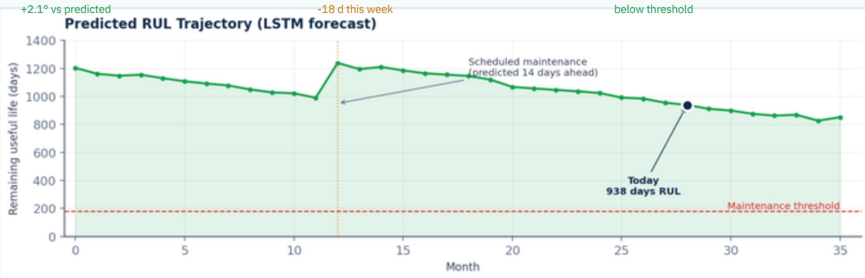
| HOTSPOT TEMP | RUL FORECAST | ANOMALY |
|---|---|---|
| 87.4°C | 412 d | 0.34 |
| +2.1° vs predicted | -18 d this week | below threshold |




**ACTIVE ALERTS & RECOMMENDATIONS**

- DGA gas pattern consistent with thermal fault <300°C — inspect cooling fans within 7 days — conf 87%
- Maintenance window scheduled: 2026-06-14 (14 days ahead of predicted threshold) — auto
- What-if simulation: pushing to 120% load for 4h would consume ~6.2 days of life — sim

## Business value

**20–40%**
Reduction in unplanned downtime
*condition-based maintenance vs schedule-based*

**15–25%**
Deferred maintenance capex
*via accurate RUL-driven scheduling*

**₹5–50 Cr**
Per catastrophic failure prevented
*replacement + outage + revenue loss avoided*

**Scalability →** The architecture generalizes from transformers to **generators, exciters, and large motors**: swap the physics module to the corresponding equipment model — the edge pipeline, data layer, ML layer, and operator dashboard are asset-agnostic. A single deployment template scales horizontally across an ABB customer's entire substation or plant fleet.